Question 1:

Part 1:

```
bodike@bodike-chaithali:~/Desktop/B24CS013_Bodike Chaithali_19$ docker network ls
NETWORK ID     NAME       DRIVER    SCOPE
39b495ec910e   bridge     bridge    local
8dbf7b20a6d1   host       host      local
0de1a7a4e3ad   lab-net    bridge    local
b67496dcde20   networkA   bridge    local
207be3570a67   none       null      local
bodike@bodike-chaithali:~/Desktop/B24CS013_Bodike Chaithali_19$ docker network create --driver bridge --subnet 182.18.0.0/16  --gateway 182.18.0.1  my
net
2153d44ee3e9ea2423489c7fbac2de1aeef8071285cb350eb6390e77ff718d52
bodike@bodike-chaithali:~/Desktop/B24CS013_Bodike Chaithali_19$ docker network ls
NETWORK ID     NAME       DRIVER    SCOPE
39b495ec910e   bridge     bridge    local
8dbf7b20a6d1   host       host      local
0de1a7a4e3ad   lab-net    bridge    local
2153d44ee3e9   mynet      bridge    local
b67496dcde20   networkA   bridge    local
207be3570a67   none       null      local
bodike@bodike-chaithali:~/Desktop/B24CS013_Bodike Chaithali_19$ docker network inspect mynet
[
    {
        "Name": "mynet",
        "Id": "2153d44ee3e9ea2423489c7fbac2de1aeef8071285cb350eb6390e77ff718d52",
        "Created": "2025-11-21T19:12:57.316418554+05:30",
        "Scope": "local",
        "Driver": "bridge",
        "EnableIPv4": true,
        "EnableIPv6": false,
        "IPAM": {
            "Driver": "default",
            "Options": {},
            "Config": [
                {
                    "Subnet": "182.18.0.0/16",
                    "IPRange": "",
                    "Gateway": "182.18.0.1"
                }
            ]
        },
        "Internal": false,
        "Attachable": false,
        "Ingress": false,
        "ConfigFrom": {
            "Network": ""
        },
        "ConfigOnly": false,
        "Options": {},
        "Labels": {},
        "Containers": {},
        "Status": {
            "IPAM": {
                "Subnets": {
                    "182.18.0.0/16": {
                        "IPsInUse": 3,
                        "DynamicIPsAvailable": 65533
                    }
                }
            }
        }
    }
]
```

Part 2:

```
bodike@bodike-chaithali:~/Desktop/B24CS013_Bodike Chaithali_19$ docker pull ubuntu
Using default tag: latest
latest: Pulling from library/ubuntu
Digest: sha256:c35e29c9450151419d9448b0fd75374fec4fff364a27f176fb458d472dfc9e54
Status: Image is up to date for ubuntu:latest
docker.io/library/ubuntu:latest
bodike@bodike-chaithali:~/Desktop/B24CS013_Bodike Chaithali_19$ docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
bodike@bodike-chaithali:~/Desktop/B24CS013_Bodike Chaithali_19$ docker run -it -d --name ubuntu1 --network mynet --ip 182.18.0.10 ubuntu bash
8813672c8fba30e8955b1a1f7200965f5be5fe67fe8de7293a1cbfb6271d543c
bodike@bodike-chaithali:~/Desktop/B24CS013_Bodike Chaithali_19$ docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED         STATUS         PORTS     NAMES
8813672c8fba   ubuntu    "bash"    7 seconds ago   Up 6 seconds             ubuntu1
bodike@bodike-chaithali:~/Desktop/B24CS013_Bodike Chaithali_19$ docker run -it -d --name ubuntu2  --network mynet --ip 182.18.0.11 ubuntu bash
5de4f912f3e09c2bc87785f2dfcc5bf19aff592b5f1225c3fddf4548cb445063
bodike@bodike-chaithali:~/Desktop/B24CS013_Bodike Chaithali_19$ docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED              STATUS              PORTS     NAMES
5de4f912f3e0   ubuntu    "bash"    32 seconds ago       Up 31 seconds               ubuntu2
8813672c8fba   ubuntu    "bash"    About a minute ago   Up About a minute           ubuntu1
bodike@bodike-chaithali:~/Desktop/B24CS013_Bodike Chaithali_19$ docker run -it -d --name ubuntu3 --network mynet ubuntu bash
5d1aacce87614ef027159b6f48853a06ace36f0b36f413f61a5583861c782224
bodike@bodike-chaithali:~/Desktop/B24CS013_Bodike Chaithali_19$ docker run -it -d --name ubuntu4 --network mynet ubuntu bash
f90142d109289eeb87e99e3567779615c8da8c29276ae12e589c5e88a094ceb1
bodike@bodike-chaithali:~/Desktop/B24CS013_Bodike Chaithali_19$ docker run -it -d --name ubuntu5 --network mynet ubuntu bash
a0b1c945634e21ecc1e1b89996bfac9f275f48a66c3ae7e3bb03ef2d1e32e19d
bodike@bodike-chaithali:~/Desktop/B24CS013_Bodike Chaithali_19$ docker run -it -d --name ubuntu6 --network mynet ubuntu bash
56e4d7d3931896cbd5c5ce7263ccc30cf4b3463287c8ca71aa69a02463aa80e3
bodike@bodike-chaithali:~/Desktop/B24CS013_Bodike Chaithali_19$ docker run -it -d --name ubuntu7 --network mynet ubuntu bash
f72b6211bd45496e99ccb4382c2a9fde3e801b6335491d453c4e3ff6fa8ca162
bodike@bodike-chaithali:~/Desktop/B24CS013_Bodike Chaithali_19$ docker run -it -d --name ubuntu8 --network mynet ubuntu bash
da7a8b361d7f4a4dfe5df893191f53f0a83f2a5f1d95f619bd49d680130914f7
bodike@bodike-chaithali:~/Desktop/B24CS013_Bodike Chaithali_19$ docker run -it -d --name ubuntu9 --network mynet ubuntu bash
58e8f84e22ae4757c8ec80aadf17f9430c7ea3d24fbbbed559b7fe3bb63cfe0f
bodike@bodike-chaithali:~/Desktop/B24CS013_Bodike Chaithali_19$ docker run -it -d --name ubuntu10 --network mynet ubuntu bash
00f1b2f875e684edd522df47b93a16b6c28e9397457aa60be8b1ab83fa811166
bodike@bodike-chaithali:~/Desktop/B24CS013_Bodike Chaithali_19$ docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED              STATUS              PORTS     NAMES
00f1b2f875e6   ubuntu    "bash"    16 seconds ago       Up 16 seconds               ubuntu10
```

```
CONTAINER ID   IMAGE     COMMAND   CREATED              STATUS              PORTS     NAMES
00f1b2f875e6   ubuntu    "bash"    16 seconds ago       Up 16 seconds               ubuntu10
58e8f84e22ae   ubuntu    "bash"    22 seconds ago       Up 21 seconds               ubuntu9
da7a8b361d7f   ubuntu    "bash"    28 seconds ago       Up 27 seconds               ubuntu8
f72b6211bd45   ubuntu    "bash"    34 seconds ago       Up 33 seconds               ubuntu7
56e4d7d39318   ubuntu    "bash"    39 seconds ago       Up 38 seconds               ubuntu6
a0b1c945634e   ubuntu    "bash"    43 seconds ago       Up 42 seconds               ubuntu5
f90142d10928   ubuntu    "bash"    47 seconds ago       Up 46 seconds               ubuntu4
5d1aacce8761   ubuntu    "bash"    52 seconds ago       Up 51 seconds               ubuntu3
5de4f912f3e0   ubuntu    "bash"    About a minute ago   Up About a minute           ubuntu2
8813672c8fba   ubuntu    "bash"    2 minutes ago        Up 2 minutes                ubuntu1
bodike@bodike-chaithali:~/Desktop/B24CS013_Bodike Chaithali_19$ docker ps -a
CONTAINER ID   IMAGE          COMMAND           CREATED              STATUS                      PORTS     NAMES
00f1b2f875e6   ubuntu         "bash"            36 seconds ago       Up 35 seconds                         ubuntu10
58e8f84e22ae   ubuntu         "bash"            42 seconds ago       Up 41 seconds                         ubuntu9
da7a8b361d7f   ubuntu         "bash"            48 seconds ago       Up 47 seconds                         ubuntu8
f72b6211bd45   ubuntu         "bash"            54 seconds ago       Up 53 seconds                         ubuntu7
56e4d7d39318   ubuntu         "bash"            59 seconds ago       Up 58 seconds                         ubuntu6
a0b1c945634e   ubuntu         "bash"            About a minute ago   Up About a minute                     ubuntu5
f90142d10928   ubuntu         "bash"            About a minute ago   Up About a minute                     ubuntu4
5d1aacce8761   ubuntu         "bash"            About a minute ago   Up About a minute                     ubuntu3
5de4f912f3e0   ubuntu         "bash"            2 minutes ago        Up 2 minutes                          ubuntu2
8813672c8fba   ubuntu         "bash"            2 minutes ago        Up 2 minutes                          ubuntu1
9ec972208742   ubuntu         "bash"            27 hours ago         Exited (137) 27 hours ago             container2
d64b8e3ccaa0   ubuntu         "bash"            27 hours ago         Exited (137) 27 hours ago             container1
1ada35932dfc   myimage:1.0    "bash start.sh"   3 days ago           Exited (0) 3 days ago                 q3-container
acbe0800fa2d   ubuntu:22.04   "sleep 600"       3 days ago           Exited (0) 3 days ago                 lab18-ubuntu-bg
3676f5628be1   ubuntu:22.04   "bash"            3 days ago           Exited (0) 3 days ago                 lab18-ubuntu
297b64c78bcc   f7931603f70e   "/hello"          3 days ago           Exited (0) 3 days ago                 happy_driscoll
bodike@bodike-chaithali:~/Desktop/B24CS013_Bodike Chaithali_19$ docker network inspect mynet
[
    {
        "Name": "mynet",
        "Id": "2153d44ee3e9ea2423489c7fbac2de1aeef8071285cb350eb6390e77ff718d52",
        "Created": "2025-11-21T19:12:57.316418554+05:30",
        "Scope": "local",
        "Driver": "bridge",
        "EnableIPv4": true,
```

```
    "EnableIPv6": false,
    "IPAM": {
        "Driver": "default",
        "Options": {},
        "Config": [
            {
                "Subnet": "182.18.0.0/16",
                "IPRange": "",
                "Gateway": "182.18.0.1"
            }
        ]
    },
    "Internal": false,
    "Attachable": false,
    "Ingress": false,
    "ConfigFrom": {
        "Network": ""
    },
    "ConfigOnly": false,
    "Options": {},
    "Labels": {},
    "Containers": {
        "00f1b2f875e684edd522df47b93a16b6c28e9397457aa60be8b1ab83fa811166": {
            "Name": "ubuntu10",
            "EndpointID": "782fe1d064c1b86501d13c3a346508afde2e8c64aa7ca5e58596ae62544dbc18",
            "MacAddress": "52:3a:1c:6f:08:f0",
            "IPv4Address": "182.18.0.9/16",
            "IPv6Address": ""
        },
        "56e4d7d3931896cbd5c5ce7263ccc30cf4b3463287c8ca71aa69a02463aa80e3": {
            "Name": "ubuntu6",
            "EndpointID": "25023bd8794570354cc1568e47bf1e21eaa8662b6aee140206aef09ae297f739",
            "MacAddress": "de:d0:6e:29:d6:0b",
            "IPv4Address": "182.18.0.5/16",
            "IPv6Address": ""
        },
        "58e8f84e22ae4757c8ec80aadf17f9430c7ea3d24fbbbed559b7fe3bb63cfe0f": {
            "Name": "ubuntu9",
            "EndpointID": "271ad78457b772f2ad481b736aa8992e6fd81c5e7e14ca99b513c23becef4f6f",
            "MacAddress": "ba:49:31:8c:2c:66",
            "IPv4Address": "182.18.0.8/16",
            "IPv6Address": ""
        },
        "5d1aacce87614ef027159b6f48853a06ace36f0b36f413f61a5583861c782224": {
            "Name": "ubuntu3",
            "EndpointID": "02e359e0adc03e6cc3e70304a11d6a68c43c11a9ef3def1e32ea46f96d19a09c",
            "MacAddress": "2e:e0:37:9e:9b:af",
            "IPv4Address": "182.18.0.2/16",
            "IPv6Address": ""
        },
        "5de4f912f3e09c2bc87785f2dfcc5bf19aff592b5f1225c3fddf4548cb445063": {
            "Name": "ubuntu2",
            "EndpointID": "a9bd7189f30f373442ba27377c0ba3e8fdba6b15ea6b713558188e3c03f03ef6",
            "MacAddress": "7e:b9:02:ba:68:2e",
            "IPv4Address": "182.18.0.11/16",
            "IPv6Address": ""
        },
        "8813672c8fba30e8955b1a1f7200965f5be5fe67fe8de7293a1cbfb6271d543c": {
            "Name": "ubuntu1",
            "EndpointID": "c470adfd21f3fb3871ea09301aefbe592abcdd44e7389d29bf843f1c21fe1ba6",
            "MacAddress": "26:bd:f6:44:d3:c3",
            "IPv4Address": "182.18.0.10/16",
            "IPv6Address": ""
        },
        "a0b1c945634e21ecc1e1b89996bfac9f275f48a66c3ae7e3bb03ef2d1e32e19d": {
            "Name": "ubuntu5",
            "EndpointID": "a157b4e7ffe9259361c821284b6223e9e40de77f92140818324c846df944ca0d",
            "MacAddress": "b2:c7:b1:1c:6e:0e",
            "IPv4Address": "182.18.0.4/16",
            "IPv6Address": ""
        },
        "da7a8b361d7f4a4dfe5df893191f53f0a83f2a5f1d95f619bd49d680130914f7": {
            "Name": "ubuntu8",
            "EndpointID": "f1af59e6ea73f6b86b94d93b9e8b62c14da125d517b91bbd629116014f5625ef",
            "MacAddress": "5a:f8:21:bb:97:96",
            "IPv4Address": "182.18.0.7/16",
            "IPv6Address": ""
```

```
            },
            "f72b6211bd45496e99ccb4382c2a9fde3e801b6335491d453c4e3ff6fa8ca162": {
                "Name": "ubuntu7",
                "EndpointID": "7a5456966dda325b7f7f4d6ffae276a9f19b511e0e7bf2fad9897beae329dabe",
                "MacAddress": "96:5c:cc:5e:0e:50",
                "IPv4Address": "182.18.0.6/16",
                "IPv6Address": ""
            },
            "f90142d109289eeb87e99e3567779615c8da8c29276ae12e589c5e88a094ceb1": {
                "Name": "ubuntu4",
                "EndpointID": "9ec5efbf4e9c2af0a97b9f0c69c2d1e851684069cbbd41a71e93228b5c550394",
                "MacAddress": "be:62:f7:42:88:0d",
                "IPv4Address": "182.18.0.3/16",
                "IPv6Address": ""
            }
        },
        "Status": {
            "IPAM": {
                "Subnets": {
                    "182.18.0.0/16": {
                        "IPsInUse": 13,
                        "DynamicIPsAvailable": 65523
                    }
                }
            }
        }
    }
]
bodike@bodike-chaithali:~/Desktop/B24CS013_Bodike Chaithali_19$
```


Part 3:

```
bodike@bodike-chaithali:~/Desktop/B24CS013_Bodike Chaithali_19$ docker exec -it ubuntu1 bash
root@8813672c8fba:/# apt update
Get:1 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble InRelease [256 kB]
Get:3 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Packages [33.1 kB]
Get:4 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [2732 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:7 http://archive.ubuntu.com/ubuntu noble/multiverse amd64 Packages [331 kB]
Get:8 http://archive.ubuntu.com/ubuntu noble/restricted amd64 Packages [117 kB]
Get:9 http://archive.ubuntu.com/ubuntu noble/main amd64 Packages [1808 kB]
Get:10 http://archive.ubuntu.com/ubuntu noble/universe amd64 Packages [19.3 MB]
Get:11 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1174 kB]
Get:12 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1659 kB]
Get:13 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [2050 kB]
Get:14 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Packages [35.9 kB]
Get:15 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [2925 kB]
Get:16 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1942 kB]
Get:17 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 Packages [49.4 kB]
Get:18 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Packages [33.9 kB]
Fetched 34.8 MB in 35s (994 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
root@8813672c8fba:/# apt install -y iputils-ping
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libcap2-bin libpam-cap
The following NEW packages will be installed:
  iputils-ping libcap2-bin libpam-cap
0 upgraded, 3 newly installed, 0 to remove and 0 not upgraded.
Need to get 91.2 kB of archives.
After this operation, 322 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libcap2-bin amd64 1:2.66-5ubuntu2.2 [34.2 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 iputils-ping amd64 3:20240117-1ubuntu0.1 [44.6 kB]
```

```
Get:3 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libpam-cap amd64 1:2.66-5ubuntu2.2 [12.5 kB]
Fetched 91.2 kB in 3s (36.3 kB/s)
debconf: delaying package configuration, since apt-utils is not installed
Selecting previously unselected package libcap2-bin.
(Reading database ... 4381 files and directories currently installed.)
Preparing to unpack .../libcap2-bin_1%3a2.66-5ubuntu2.2_amd64.deb ...
Unpacking libcap2-bin (1:2.66-5ubuntu2.2) ...
Selecting previously unselected package iputils-ping.
Preparing to unpack .../iputils-ping_3%3a20240117-1ubuntu0.1_amd64.deb ...
Unpacking iputils-ping (3:20240117-1ubuntu0.1) ...
Selecting previously unselected package libpam-cap:amd64.
Preparing to unpack .../libpam-cap_1%3a2.66-5ubuntu2.2_amd64.deb ...
Unpacking libpam-cap:amd64 (1:2.66-5ubuntu2.2) ...
Setting up libcap2-bin (1:2.66-5ubuntu2.2) ...
Setting up libpam-cap:amd64 (1:2.66-5ubuntu2.2) ...
debconf: unable to initialize frontend: Dialog
debconf: (No usable dialog-like program is installed, so the dialog based frontend cannot be used. at /usr/share/perl5/Debconf/FrontEnd/Dialog.pm line
 79.)
debconf: falling back to frontend: Readline
debconf: unable to initialize frontend: Readline
debconf: (Can't locate Term/ReadLine.pm in @INC (you may need to install the Term::ReadLine module) (@INC entries checked: /etc/perl /usr/local/lib/x8
6_64-linux-gnu/perl/5.38.2 /usr/local/share/perl/5.38.2 /usr/lib/x86_64-linux-gnu/perl5/5.38 /usr/share/perl5 /usr/lib/x86_64-linux-gnu/perl-base /usr
/lib/x86_64-linux-gnu/perl/5.38 /usr/share/perl/5.38 /usr/local/lib/site_perl) at /usr/share/perl5/Debconf/FrontEnd/Readline.pm line 8.)
debconf: falling back to frontend: Teletype
Setting up iputils-ping (3:20240117-1ubuntu0.1) ...
root@8813672c8fba:/# exit
exit
bodike@bodike-chaithali:~/Desktop/B24CS013_Bodike Chaithali_19$ docker exec -it ubuntu5 bash
root@a0b1c945634e:/# apt update
Get:1 http://archive.ubuntu.com/ubuntu noble InRelease [256 kB]
Get:2 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:3 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1174 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble/multiverse amd64 Packages [331 kB]
Get:7 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [2732 kB]
Get:8 http://archive.ubuntu.com/ubuntu noble/main amd64 Packages [1808 kB]
```

```
Get:9 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Packages [33.1 kB]
Get:10 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1659 kB]
Get:11 http://archive.ubuntu.com/ubuntu noble/restricted amd64 Packages [117 kB]
Get:12 http://archive.ubuntu.com/ubuntu noble/universe amd64 Packages [19.3 MB]
Get:13 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [2050 kB]
Get:14 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Packages [35.9 kB]
Get:15 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1942 kB]
Get:16 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [2925 kB]
Get:17 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Packages [33.9 kB]
Get:18 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 Packages [49.4 kB]
Fetched 34.8 MB in 10s (3478 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
root@a0b1c945634e:/# apt install -y iputils-ping
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libcap2-bin libpam-cap
The following NEW packages will be installed:
  iputils-ping libcap2-bin libpam-cap
0 upgraded, 3 newly installed, 0 to remove and 0 not upgraded.
Need to get 91.2 kB of archives.
After this operation, 322 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libcap2-bin amd64 1:2.66-5ubuntu2.2 [34.2 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 iputils-ping amd64 3:20240117-1ubuntu0.1 [44.6 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libpam-cap amd64 1:2.66-5ubuntu2.2 [12.5 kB]
Fetched 91.2 kB in 5s (18.8 kB/s)
debconf: delaying package configuration, since apt-utils is not installed
Selecting previously unselected package libcap2-bin.
(Reading database ... 4381 files and directories currently installed.)
Preparing to unpack .../libcap2-bin_1%3a2.66-5ubuntu2.2_amd64.deb ...
Unpacking libcap2-bin (1:2.66-5ubuntu2.2) ...
Selecting previously unselected package iputils-ping.
Preparing to unpack .../iputils-ping_3%3a20240117-1ubuntu0.1_amd64.deb ...
Unpacking iputils-ping (3:20240117-1ubuntu0.1) ...
```

```
Unpacking iputils-ping (3:20240117-1ubuntu0.1) ...
Selecting previously unselected package libpam-cap:amd64.
Preparing to unpack .../libpam-cap_1%3a2.66-5ubuntu2.2_amd64.deb ...
Unpacking libpam-cap:amd64 (1:2.66-5ubuntu2.2) ...
Setting up libcap2-bin (1:2.66-5ubuntu2.2) ...
Setting up libpam-cap:amd64 (1:2.66-5ubuntu2.2) ...
debconf: unable to initialize frontend: Dialog
debconf: (No usable dialog-like program is installed, so the dialog based frontend cannot be used. at /usr/share/perl5/Debconf/FrontEnd/Dialog.pm line
 79.)
debconf: falling back to frontend: Readline
debconf: unable to initialize frontend: Readline
debconf: (Can't locate Term/ReadLine.pm in @INC (you may need to install the Term::ReadLine module) (@INC entries checked: /etc/perl /usr/local/lib/x8
6_64-linux-gnu/perl/5.38.2 /usr/local/share/perl/5.38.2 /usr/lib/x86_64-linux-gnu/perl5/5.38 /usr/share/perl5 /usr/lib/x86_64-linux-gnu/perl-base /usr
/lib/x86_64-linux-gnu/perl/5.38 /usr/share/perl/5.38 /usr/local/lib/site_perl) at /usr/share/perl5/Debconf/FrontEnd/Readline.pm line 8.)
debconf: falling back to frontend: Teletype
Setting up iputils-ping (3:20240117-1ubuntu0.1) ...
root@a0b1c945634e:/# exit
exit
bodike@bodike-chaithali:~/Desktop/B24CS013_Bodike Chaithali_19$
```

Part 4:

```
bodike@bodike-chaithali:~/Desktop/B24CS013_Bodike Chaithali_19$ docker exec -it ubuntu1 bash
root@8813672c8fba:/# ping -c 3 ubuntu4
PING ubuntu4 (182.18.0.3) 56(84) bytes of data.
64 bytes from ubuntu4.mynet (182.18.0.3): icmp_seq=1 ttl=64 time=0.252 ms
64 bytes from ubuntu4.mynet (182.18.0.3): icmp_seq=2 ttl=64 time=0.132 ms
64 bytes from ubuntu4.mynet (182.18.0.3): icmp_seq=3 ttl=64 time=0.114 ms

--- ubuntu4 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2053ms
rtt min/avg/max/mdev = 0.114/0.166/0.252/0.061 ms
root@8813672c8fba:/# ping -c 3 ubuntu7
PING ubuntu7 (182.18.0.6) 56(84) bytes of data.
64 bytes from ubuntu7.mynet (182.18.0.6): icmp_seq=1 ttl=64 time=0.198 ms
64 bytes from ubuntu7.mynet (182.18.0.6): icmp_seq=2 ttl=64 time=0.119 ms
64 bytes from ubuntu7.mynet (182.18.0.6): icmp_seq=3 ttl=64 time=0.115 ms

--- ubuntu7 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2063ms
rtt min/avg/max/mdev = 0.115/0.144/0.198/0.038 ms
root@8813672c8fba:/# ping -c 3 ubuntu9
PING ubuntu9 (182.18.0.8) 56(84) bytes of data.
64 bytes from ubuntu9.mynet (182.18.0.8): icmp_seq=1 ttl=64 time=0.167 ms
64 bytes from ubuntu9.mynet (182.18.0.8): icmp_seq=2 ttl=64 time=0.122 ms
64 bytes from ubuntu9.mynet (182.18.0.8): icmp_seq=3 ttl=64 time=0.120 ms

--- ubuntu9 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2086ms
rtt min/avg/max/mdev = 0.120/0.136/0.167/0.021 ms
root@8813672c8fba:/# ping -c 3 ubuntu10
PING ubuntu10 (182.18.0.9) 56(84) bytes of data.
64 bytes from ubuntu10.mynet (182.18.0.9): icmp_seq=1 ttl=64 time=0.151 ms
64 bytes from ubuntu10.mynet (182.18.0.9): icmp_seq=2 ttl=64 time=0.117 ms
64 bytes from ubuntu10.mynet (182.18.0.9): icmp_seq=3 ttl=64 time=0.110 ms

--- ubuntu10 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2053ms
rtt min/avg/max/mdev = 0.110/0.126/0.151/0.017 ms
root@8813672c8fba:/# exit
```

```
exit
bodike@bodike-chaithali:~/Desktop/B24CS013_Bodike Chaithali_19$ docker network inspect mynet
[
    {
        "Name": "mynet",
        "Id": "2153d44ee3e9ea2423489c7fbac2de1aeef8071285cb350eb6390e77ff718d52",
        "Created": "2025-11-21T19:12:57.316418554+05:30",
        "Scope": "local",
        "Driver": "bridge",
        "EnableIPv4": true,
        "EnableIPv6": false,
        "IPAM": {
            "Driver": "default",
            "Options": {},
            "Config": [
                {
                    "Subnet": "182.18.0.0/16",
                    "IPRange": "",
                    "Gateway": "182.18.0.1"
                }
            ]
        },
        "Internal": false,
        "Attachable": false,
        "Ingress": false,
        "ConfigFrom": {
            "Network": ""
        },
        "ConfigOnly": false,
        "Options": {},
        "Labels": {},
        "Containers": {
            "00f1b2f875e684edd522df47b93a16b6c28e9397457aa60be8b1ab83fa811166": {
                "Name": "ubuntu10",
                "EndpointID": "782fe1d064c1b86501d13c3a346508afde2e8c64aa7ca5e58596ae62544dbc18",
                "MacAddress": "52:3a:1c:6f:08:f0",
                "IPv4Address": "182.18.0.9/16",
                "IPv6Address": ""
```

```
        },
        "56e4d7d3931896cbd5c5ce7263ccc30cf4b3463287c8ca71aa69a02463aa80e3": {
            "Name": "ubuntu6",
            "EndpointID": "25023bd8794570354cc1568e47bf1e21eaa8662b6aee140206aef09ae297f739",
            "MacAddress": "de:d0:6e:29:d6:0b",
            "IPv4Address": "182.18.0.5/16",
            "IPv6Address": ""
        },
        "58e8f84e22ae4757c8ec80aadf17f9430c7ea3d24fbbbed559b7fe3bb63cfe0f": {
            "Name": "ubuntu9",
            "EndpointID": "271ad78457b772f2ad481b736aa8992e6fd81c5e7e14ca99b513c23becef4f6f",
            "MacAddress": "ba:49:31:8c:2c:66",
            "IPv4Address": "182.18.0.8/16",
            "IPv6Address": ""
        },
        "5d1aacce87614ef027159b6f48853a06ace36f0b36f413f61a5583861c782224": {
            "Name": "ubuntu3",
            "EndpointID": "02e359e0adc03e6cc3e70304a11d6a68c43c11a9ef3def1e32ea46f96d19a09c",
            "MacAddress": "2e:e0:37:9e:9b:af",
            "IPv4Address": "182.18.0.2/16",
            "IPv6Address": ""
        },
        "5de4f912f3e09c2bc87785f2dfcc5bf19aff592b5f1225c3fddf4548cb445063": {
            "Name": "ubuntu2",
            "EndpointID": "a9bd7189f30f373442ba27377c0ba3e8fdba6b15ea6b713558188e3c03f03ef6",
            "MacAddress": "7e:b9:02:ba:68:2e",
            "IPv4Address": "182.18.0.11/16",
            "IPv6Address": ""
        },
        "8813672c8fba30e8955b1a1f7200965f5be5fe67fe8de7293a1cbfb6271d543c": {
            "Name": "ubuntu1",
            "EndpointID": "c470adfd21f3fb3871ea09301aefbe592abcdd44e7389d29bf843f1c21fe1ba6",
            "MacAddress": "26:bd:f6:44:d3:c3",
            "IPv4Address": "182.18.0.10/16",
            "IPv6Address": ""
        },
        "a0b1c945634e21ecc1e1b89996bfac9f275f48a66c3ae7e3bb03ef2d1e32e19d": {
            "Name": "ubuntu5",
```



```
            "EndpointID": "a157b4e7ffe9259361c821284b6223e9e40de77f92140818324c846df944ca0d",
            "MacAddress": "b2:c7:b1:1c:6e:0e",
            "IPv4Address": "182.18.0.4/16",
            "IPv6Address": ""
        },
        "da7a8b361d7f4a4dfe5df893191f53f0a83f2a5f1d95f619bd49d680130914f7": {
            "Name": "ubuntu8",
            "EndpointID": "f1af59e6ea73f6b86b94d93b9e8b62c14da125d517b91bbd629116014f5625ef",
            "MacAddress": "5a:f8:21:bb:97:96",
            "IPv4Address": "182.18.0.7/16",
            "IPv6Address": ""
        },
        "f72b6211bd45496e99ccb4382c2a9fde3e801b6335491d453c4e3ff6fa8ca162": {
            "Name": "ubuntu7",
            "EndpointID": "7a5456966dda325b7f7f4d6ffae276a9f19b511e0e7bf2fad9897beae329dabe",
            "MacAddress": "96:5c:cc:5e:0e:50",
            "IPv4Address": "182.18.0.6/16",
            "IPv6Address": ""
        },
        "f90142d109289eeb87e99e3567779615c8da8c29276ae12e589c5e88a094ceb1": {
            "Name": "ubuntu4",
            "EndpointID": "9ec5efbf4e9c2af0a97b9f0c69c2d1e851684069cbbd41a71e93228b5c550394",
            "MacAddress": "be:62:f7:42:88:0d",
            "IPv4Address": "182.18.0.3/16",
            "IPv6Address": ""
        }
    },
    "Status": {
        "IPAM": {
            "Subnets": {
                "182.18.0.0/16": {
                    "IPsInUse": 13,
                    "DynamicIPsAvailable": 65523
                }
            }
        }
    }
}
]
```

```
bodike@bodike-chaithali:~/Desktop/B24CS013_Bodike Chaithali_19$ docker exec -it ubuntu5 bash
root@a0b1c945634e:/# ping -c 3 182.18.0.23
PING 182.18.0.23 (182.18.0.23) 56(84) bytes of data.
From 182.18.0.4 icmp_seq=1 Destination Host Unreachable
From 182.18.0.4 icmp_seq=2 Destination Host Unreachable
From 182.18.0.4 icmp_seq=3 Destination Host Unreachable

--- 182.18.0.23 ping statistics ---
3 packets transmitted, 0 received, +3 errors, 100% packet loss, time 2041ms
pipe 3
root@a0b1c945634e:/# ping -c 3 182.18.0.25
PING 182.18.0.25 (182.18.0.25) 56(84) bytes of data.
From 182.18.0.4 icmp_seq=1 Destination Host Unreachable
From 182.18.0.4 icmp_seq=2 Destination Host Unreachable
From 182.18.0.4 icmp_seq=3 Destination Host Unreachable

--- 182.18.0.25 ping statistics ---
3 packets transmitted, 0 received, +3 errors, 100% packet loss, time 2081ms
pipe 3
root@a0b1c945634e:/# ping -c 3 182.18.0.32
PING 182.18.0.32 (182.18.0.32) 56(84) bytes of data.
From 182.18.0.4 icmp_seq=1 Destination Host Unreachable
From 182.18.0.4 icmp_seq=2 Destination Host Unreachable
From 182.18.0.4 icmp_seq=3 Destination Host Unreachable

--- 182.18.0.32 ping statistics ---
3 packets transmitted, 0 received, +3 errors, 100% packet loss, time 2064ms
pipe 3
root@a0b1c945634e:/# ping -c 3 182.18.0.40
PING 182.18.0.40 (182.18.0.40) 56(84) bytes of data.
From 182.18.0.4 icmp_seq=1 Destination Host Unreachable
From 182.18.0.4 icmp_seq=2 Destination Host Unreachable
From 182.18.0.4 icmp_seq=3 Destination Host Unreachable

--- 182.18.0.40 ping statistics ---
3 packets transmitted, 0 received, +3 errors, 100% packet loss, time 2086ms
pipe 3
root@a0b1c945634e:/# exit
exit
```

Part 5:

```
root@a0b1c945634e:/# exit
exit
bodike@bodike-chaithali:~/Desktop/B24CS013_Bodike Chaithali_19$ docker network inspect mynet
[
    {
        "Name": "mynet",
        "Id": "2153d44ee3e9ea2423489c7fbac2de1aeef8071285cb350eb6390e77ff718d52",
        "Created": "2025-11-21T19:12:57.316418554+05:30",
        "Scope": "local",
        "Driver": "bridge",
        "EnableIPv4": true,
        "EnableIPv6": false,
        "IPAM": {
            "Driver": "default",
            "Options": {},
            "Config": [
                {
                    "Subnet": "182.18.0.0/16",
                    "IPRange": "",
                    "Gateway": "182.18.0.1"
                }
            ]
        },
        "Internal": false,
        "Attachable": false,
        "Ingress": false,
        "ConfigFrom": {
            "Network": ""
        },
        "ConfigOnly": false,
        "Options": {},
        "Labels": {},
        "Containers": {
            "00f1b2f875e684edd522df47b93a16b6c28e9397457aa60be8b1ab83fa811166": {
                "Name": "ubuntu10",
                "EndpointID": "782fe1d064c1b86501d13c3a346508afde2e8c64aa7ca5e58596ae62544dbc18",
                "MacAddress": "52:3a:1c:6f:08:f0",
                "IPv4Address": "182.18.0.9/16",
                "IPv6Address": ""
```

```
            "IPv4Address": "182.18.0.9/16",
            "IPv6Address": ""
        },
        "56e4d7d3931896cbd5c5ce7263ccc30cf4b3463287c8ca71aa69a02463aa80e3": {
            "Name": "ubuntu6",
            "EndpointID": "25023bd8794570354cc1568e47bf1e21eaa8662b6aee140206aef09ae297f739",
            "MacAddress": "de:d0:6e:29:d6:0b",
            "IPv4Address": "182.18.0.5/16",
            "IPv6Address": ""
        },
        "58e8f84e22ae4757c8ec80aadf17f9430c7ea3d24fbbbed559b7fe3bb63cfe0f": {
            "Name": "ubuntu9",
            "EndpointID": "271ad78457b772f2ad481b736aa8992e6fd81c5e7e14ca99b513c23becef4f6f",
            "MacAddress": "ba:49:31:8c:2c:66",
            "IPv4Address": "182.18.0.8/16",
            "IPv6Address": ""
        },
        "5d1aacce87614ef027159b6f48853a06ace36f0b36f413f61a5583861c782224": {
            "Name": "ubuntu3",
            "EndpointID": "02e359e0adc03e6cc3e70304a11d6a68c43c11a9ef3def1e32ea46f96d19a09c",
            "MacAddress": "2e:e0:37:9e:9b:af",
            "IPv4Address": "182.18.0.2/16",
            "IPv6Address": ""
        },
        "5de4f912f3e09c2bc87785f2dfcc5bf19aff592b5f1225c3fddf4548cb445063": {
            "Name": "ubuntu2",
            "EndpointID": "a9bd7189f30f373442ba27377c0ba3e8fdba6b15ea6b713558188e3c03f03ef6",
            "MacAddress": "7e:b9:02:ba:68:2e",
            "IPv4Address": "182.18.0.11/16",
            "IPv6Address": ""
        },
        "8813672c8fba30e8955b1a1f7200965f5be5fe67fe8de7293a1cbfb6271d543c": {
            "Name": "ubuntu1",
            "EndpointID": "c470adfd21f3fb3871ea09301aefbe592abcdd44e7389d29bf843f1c21fe1ba6",
            "MacAddress": "26:bd:f6:44:d3:c3",
            "IPv4Address": "182.18.0.10/16",
            "IPv6Address": ""
        },
```



```
        },
        "a0b1c945634e21ecc1e1b89996bfac9f275f48a66c3ae7e3bb03ef2d1e32e19d": {
            "Name": "ubuntu5",
            "EndpointID": "a157b4e7ffe9259361c821284b6223e9e40de77f92140818324c846df944ca0d",
            "MacAddress": "b2:c7:b1:1c:6e:0e",
            "IPv4Address": "182.18.0.4/16",
            "IPv6Address": ""
        },
        "da7a8b361d7f4a4dfe5df893191f53f0a83f2a5f1d95f619bd49d680130914f7": {
            "Name": "ubuntu8",
            "EndpointID": "f1af59e6ea73f6b86b94d93b9e8b62c14da125d517b91bbd629116014f5625ef",
            "MacAddress": "5a:f8:21:bb:97:96",
            "IPv4Address": "182.18.0.7/16",
            "IPv6Address": ""
        },
        "f72b6211bd45496e99ccb4382c2a9fde3e801b6335491d453c4e3ff6fa8ca162": {
            "Name": "ubuntu7",
            "EndpointID": "7a5456966dda325b7f7f4d6ffae276a9f19b511e0e7bf2fad9897beae329dabe",
            "MacAddress": "96:5c:cc:5e:0e:50",
            "IPv4Address": "182.18.0.6/16",
            "IPv6Address": ""
        },
        "f90142d109289eeb87e99e3567779615c8da8c29276ae12e589c5e88a094ceb1": {
            "Name": "ubuntu4",
            "EndpointID": "9ec5efbf4e9c2af0a97b9f0c69c2d1e851684069cbbd41a71e93228b5c550394",
            "MacAddress": "be:62:f7:42:88:0d",
            "IPv4Address": "182.18.0.3/16",
            "IPv6Address": ""
        }
    },
    "Status": {
        "IPAM": {
            "Subnets": {
                "182.18.0.0/16": {
                    "IPsInUse": 13,
                    "DynamicIPsAvailable": 65523
                }
            }
        }
```


1) ubuntu1 → 182.18.0.10
   ubuntu2 → 182.18.0.11
   ubuntu3 → 182.18.0.2
   ubuntu4 → 182.18.0.3
   ubuntu5 → 182.18.0.4
   ubuntu6 → 182.18.0.5
   ubuntu7 → 182.18.0.6
   ubuntu8 → 182.18.0.7
   ubuntu9 → 182.18.0.8
   ubuntu10 → 182.18.0.9

2) Gateway: 182.18.0.1

3) ubuntu1 MAC Address → 26:bd:f6:44:d3:c3

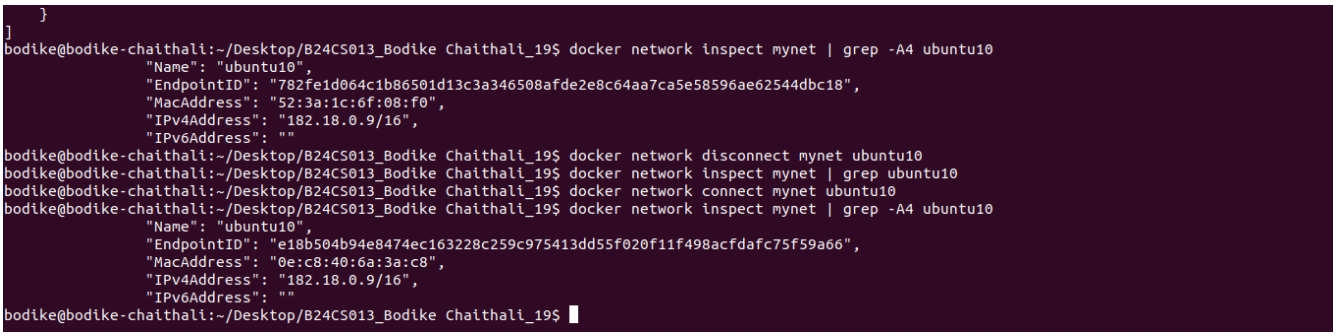
4) The Containers section lists all containers connected to the mynet network, along with:

- Container ID
- Container name
- Endpoint ID
- MAC address
- IPv4/IPv6 assigned to the container

It essentially describes how Docker attaches each container to the bridge network and shows network level configuration for each container.

5) DNS Server → 127.0.0.11 (Docker Embedded DNS)

Part 6:



Question 2:

Step 1:

```
bodike@bodike-chaithali:~/Desktop/B24CS013_Bodike Chaithali_19$ vi lab_exam.c
bodike@bodike-chaithali:~/Desktop/B24CS013_Bodike Chaithali_19$ gcc -g -Wall -Wextra lab_exam.c -o lab_exam
bodike@bodike-chaithali:~/Desktop/B24CS013_Bodike Chaithali_19$ ls -l lab_exam
-rwxrwxr-x 1 bodike bodike 17896 Nov 21 19:57 lab_exam
bodike@bodike-chaithali:~/Desktop/B24CS013_Bodike Chaithali_19$ gcc -g -Wall -Wextra lab_exam.c -o lab_exam
bodike@bodike-chaithali:~/Desktop/B24CS013_Bodike Chaithali_19$ file lab_exam
lab_exam: ELF 64-bit LSB pie executable, x86-64, version 1 (SYSV), dynamically linked, interpreter /lib64/ld-linux-x86-64.so.2, BuildID[sha1]=f8b519f7
54e7ec62cbc623d5941016fa69aa841c, for GNU/Linux 3.2.0, with debug_info, not stripped
bodike@bodike-chaithali:~/Desktop/B24CS013_Bodike Chaithali_19$ gdb ./lab_exam
GNU gdb (Ubuntu 12.1-0ubuntu1~22.04.2) 12.1
Copyright (C) 2022 Free Software Foundation, Inc.
License GPLv3+: GNU GPL version 3 or later <http://gnu.org/licenses/gpl.html>
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.
Type "show copying" and "show warranty" for details.
This GDB was configured as "x86_64-linux-gnu".
Type "show configuration" for configuration details.
For bug reporting instructions, please see:
<https://www.gnu.org/software/gdb/bugs/>.
Find the GDB manual and other documentation resources online at:
    <http://www.gnu.org/software/gdb/documentation/>.

For help, type "help".
Type "apropos word" to search for commands related to "word"...
Reading symbols from ./lab_exam...
(gdb) info functions
All defined functions:

File lab_exam.c:
19:     int main();
10:     int max_element(int *, int);
3:      int sum_array(int *, int);

Non-debugging symbols:
0x0000000000001000  _init
0x0000000000001050  __cxa_finalize@plt
0x0000000000001060  __stack_chk_fail@plt
0x0000000000001070  printf@plt
```

```
File lab_exam.c:
19:     int main();
10:     int max_element(int *, int);
3:      int sum_array(int *, int);

Non-debugging symbols:
0x0000000000001000  _init
0x0000000000001050  __cxa_finalize@plt
0x0000000000001060  __stack_chk_fail@plt
0x0000000000001070  printf@plt
0x0000000000001080  _start
0x00000000000010b0  deregister_tm_clones
0x00000000000010e0  register_tm_clones
0x0000000000001120  __do_global_dtors_aux
0x0000000000001160  frame_dummy
0x00000000000012dc  _fini
(gdb) quit
```

```
}
bodike@bodike-chaithali:~/Desktop/B24CS013_Bodike Chaithali_19$ cat lab_exam.c
#include <stdio.h>

int sum_array(int arr[], int size) {
        int sum = 0;
        for(int i = 0; i < size; i++)
                sum += arr[i];
        return sum;
}

int max_element(int arr[], int size) {
        int max = arr[0];
        for(int i = 1; i < size; i++) {
                if(arr[i] > max)
                        max = arr[i];
        }
        return max;
}

int main() {
        int data[] = {4, 8, 15, 16, 23, 42};
        int n = sizeof(data) / sizeof(data[0]);
        int total = sum_array(data, n);
        printf("Sum: %d\n", total);
        int maximum = max_element(data, n);
        printf("Max: %d\n", maximum);
        return 0;
}
bodike@bodike-chaithali:~/Desktop/B24CS013_Bodike Chaithali_19$
```

Step 2:

```
bodike@bodike-chaithali:~/Desktop/B24CS013_Bodike Chaithali_19$ ls
lab_exam  lab_exam.c
bodike@bodike-chaithali:~/Desktop/B24CS013_Bodike Chaithali_19$ gdb ./lab_exam
GNU gdb (Ubuntu 12.1-0ubuntu1~22.04.2) 12.1
Copyright (C) 2022 Free Software Foundation, Inc.
License GPLv3+: GNU GPL version 3 or later <http://gnu.org/licenses/gpl.html>
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.
Type "show copying" and "show warranty" for details.
This GDB was configured as "x86_64-linux-gnu".
Type "show configuration" for configuration details.
For bug reporting instructions, please see:
<https://www.gnu.org/software/gdb/bugs/>.
Find the GDB manual and other documentation resources online at:
    <http://www.gnu.org/software/gdb/documentation/>.

For help, type "help".
Type "apropos word" to search for commands related to "word"...
Reading symbols from ./lab_exam...
(gdb) list main
14                              max = arr[i];
15              }
16              return max;
17      }
18
19      int main() {
20              int data[] = {4, 8, 15, 16, 23, 42};
21              int n = sizeof(data) / sizeof(data[0]);
22              int total = sum_array(data, n);
23              printf("Sum: %d\n", total);
(gdb) break main
Breakpoint 1 at 0x1224: file lab_exam.c, line 19.
(gdb) run
Starting program: /home/bodike/Desktop/B24CS013_Bodike Chaithali_19/lab_exam

[Thread debugging using libthread_db enabled]
Using host libthread_db library "/lib/x86_64-linux-gnu/libthread_db.so.1".

Breakpoint 1, main () at lab_exam.c:19
19      int main() {
(gdb)
```

Step 3:

```
Breakpoint 1 at 0x1224: file lab_exam.c, line 19.
(gdb) run
Starting program: /home/bodike/Desktop/B24CS013_Bodike Chaithali_19/lab_exam

[Thread debugging using libthread_db enabled]
Using host libthread_db library "/lib/x86_64-linux-gnu/libthread_db.so.1".

Breakpoint 1, main () at lab_exam.c:19
19      int main() {
(gdb) next
20              int data[] = {4, 8, 15, 16, 23, 42};
(gdb) next
21              int n = sizeof(data) / sizeof(data[0]);
(gdb) next
22              int total = sum_array(data, n);
(gdb) step
sum_array (arr=0x7fffffffe580, size=6) at lab_exam.c:4
4               int sum = 0;
(gdb) next
5               for(int i = 0; i < size; i++)
(gdb) display i
1: i = 32767
(gdb) display sum
2: sum = 0
```

```
(gdb) next
6                       sum += arr[i];
1: i = 0
2: sum = 0
3: arr[i] = 4
(gdb) display arr[i]
4: arr[i] = 4
(gdb)
```

Step 4:

```
(gdb) finish
Run till exit from #0  sum_array (arr=0x7fffffffe580, size=6) at lab_exam.c:6
0x0000555555555275 in main () at lab_exam.c:22
22              int total = sum_array(data, n);
Value returned is $1 = 108
(gdb) print data
$2 = {4, 8, 15, 16, 23, 42}
(gdb) print data[0]@6
$3 = {4, 8, 15, 16, 23, 42}
(gdb) print n
$4 = 6
(gdb) next
23              printf("Sum: %d\n", total);
(gdb) print total
$5 = 108
(gdb) next
Sum: 108
24              int maximum = max_element(data, n);
(gdb) print maximum
$6 = 0
(gdb) next
25              printf("Max: %d\n", maximum);
(gdb) print maximum
$7 = 42
```

Step 5:

```
bodike@bodike-chaithali:~/Desktop/B24CS013_Bodike Chaithali_19$ gdb ./lab_exam
GNU gdb (Ubuntu 12.1-0ubuntu1~22.04.2) 12.1
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This GDB was configured as "x86_64-linux-gnu".
Type "show configuration" for configuration details.
For bug reporting instructions, please see:
<https://www.gnu.org/software/gdb/bugs/>.
Find the GDB manual and other documentation resources online at:
    <http://www.gnu.org/software/gdb/documentation/>.

For help, type "help".
Type "apropos word" to search for commands related to "word"...
Reading symbols from ./lab_exam...
(gdb) break sum_array
Breakpoint 1 at 0x1178: file lab_exam.c, line 4.
(gdb) run
Starting program: /home/bodike/Desktop/B24CS013_Bodike Chaithali_19/lab_exam
[Thread debugging using libthread_db enabled]
Using host libthread_db library "/lib/x86_64-linux-gnu/libthread_db.so.1".

Breakpoint 1, sum_array (arr=0x7fffffffe580, size=6) at lab_exam.c:4
4               int sum = 0;
(gdb) next
5               for(int i = 0; i < size; i++)
(gdb) next
6                       sum += arr[i];
(gdb) watch sum
Hardware watchpoint 2: sum
(gdb) next

Hardware watchpoint 2: sum
```

```
Hardware watchpoint 2: sum

Old value = 0
New value = 4
sum_array (arr=0x7fffffffe580, size=6) at lab_exam.c:5
5               for(int i = 0; i < size; i++)
(gdb) next
6                       sum += arr[i];
(gdb) watch sum
Hardware watchpoint 3: sum
(gdb) next

Hardware watchpoint 2: sum

Old value = 4
New value = 12

Hardware watchpoint 3: sum

Old value = 4
New value = 12
sum_array (arr=0x7fffffffe580, size=6) at lab_exam.c:5
5               for(int i = 0; i < size; i++)
(gdb) break 6 if arr[i] == 16
Breakpoint 4 at 0x555555555188: file lab_exam.c, line 6.
(gdb) continue
Continuing.

Hardware watchpoint 2: sum

Old value = 12
New value = 27

Hardware watchpoint 3: sum

Old value = 12
New value = 27
sum_array (arr=0x7fffffffe580, size=6) at lab_exam.c:5
```

```
Old value = 12
New value = 27

Hardware watchpoint 3: sum

Old value = 12
New value = 27
sum_array (arr=0x7fffffffe580, size=6) at lab_exam.c:5
5               for(int i = 0; i < size; i++)
(gdb)
```

Step 6:

```
bodike@bodike-chaithali:~/Desktop/B24CS013_Bodike Chaithali_19$ gdb ./lab_exam
GNU gdb (Ubuntu 12.1-0ubuntu1~22.04.2) 12.1
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Type "show configuration" for configuration details.
For bug reporting instructions, please see:
<https://www.gnu.org/software/gdb/bugs/>.
Find the GDB manual and other documentation resources online at:
    <http://www.gnu.org/software/gdb/documentation/>.

For help, type "help".
Type "apropos word" to search for commands related to "word"...
Reading symbols from ./lab_exam...
(gdb) break max_element
Breakpoint 1 at 0x11c1: file lab_exam.c, line 11.
(gdb) run
Starting program: /home/bodike/Desktop/B24CS013_Bodike Chaithali_19/lab_exam
[Thread debugging using libthread_db enabled]
Using host libthread_db library "/lib/x86_64-linux-gnu/libthread_db.so.1".
Sum: 108

Breakpoint 1, max_element (arr=0x7fffffffe580, size=6) at lab_exam.c:11
11              int max = arr[0];
(gdb) display max
1: max = 108
(gdb) next
12              for(int i = 1; i < size; i++) {
1: max = 4
(gdb) next
13                      if(arr[i] > max)
1: max = 4
(gdb) next
14                              max = arr[i];
1: max = 4
```

```
13                      if(arr[i] > max)
1: max = 4
(gdb) next
14                              max = arr[i];
1: max = 4
(gdb) next
12              for(int i = 1; i < size; i++) {
1: max = 8
(gdb) next
13                      if(arr[i] > max)
1: max = 8
(gdb) next
14                              max = arr[i];
1: max = 8
(gdb) next
12              for(int i = 1; i < size; i++) {
1: max = 15
(gdb) next
13                      if(arr[i] > max)
1: max = 15
(gdb) next
14                              max = arr[i];
1: max = 15
(gdb) next
12              for(int i = 1; i < size; i++) {
1: max = 16
(gdb) next
13                      if(arr[i] > max)
1: max = 16
(gdb) next
14                              max = arr[i];
1: max = 16
(gdb) next
12              for(int i = 1; i < size; i++) {
1: max = 23
(gdb) display arr[i]
2: arr[i] = 23
(gdb)
```

Step 7:

```
bodike@bodike-chaithali:~/Desktop/B24CS013_Bodike Chaithali_19$ gdb ./lab_exam
GNU gdb (Ubuntu 12.1-0ubuntu1~22.04.2) 12.1
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This GDB was configured as "x86_64-linux-gnu".
Type "show configuration" for configuration details.
For bug reporting instructions, please see:
<https://www.gnu.org/software/gdb/bugs/>.
Find the GDB manual and other documentation resources online at:
    <http://www.gnu.org/software/gdb/documentation/>.

For help, type "help".
Type "apropos word" to search for commands related to "word"...
Reading symbols from ./lab_exam...
(gdb) break max_element
Breakpoint 1 at 0x11c1: file lab_exam.c, line 11.
(gdb) run
Starting program: /home/bodike/Desktop/B24CS013_Bodike Chaithali_19/lab_exam
[Thread debugging using libthread_db enabled]
Using host libthread_db library "/lib/x86_64-linux-gnu/libthread_db.so.1".
Sum: 108

Breakpoint 1, max_element (arr=0x7fffffffe580, size=6) at lab_exam.c:11
11              int max = arr[0];
(gdb) backtrace
#0  max_element (arr=0x7fffffffe580, size=6) at lab_exam.c:11
#1  0x00005555555552a2 in main () at lab_exam.c:24
(gdb) frame 0
#0  max_element (arr=0x7fffffffe580, size=6) at lab_exam.c:11
11              int max = arr[0];
(gdb) up
#1  0x00005555555552a2 in main () at lab_exam.c:24
24              int maximum = max_element(data, n);
(gdb) down
#0  max_element (arr=0x7fffffffe580, size=6) at lab_exam.c:11
11              int max = arr[0];
(gdb) exit
A debugging session is active.

        Inferior 1 [process 15321] will be killed.

Quit anyway? (y or n) y
bodike@bodike-chaithali:~/Desktop/B24CS013_Bodike Chaithali_19$
```

Step 8:

```
(gdb) exit
bodike@bodike-chaithali:~/Desktop/B24CS013_Bodike Chaithali_19$ gdb ./lab_exam
GNU gdb (Ubuntu 12.1-0ubuntu1~22.04.2) 12.1
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This GDB was configured as "x86_64-linux-gnu".
Type "show configuration" for configuration details.
For bug reporting instructions, please see:
<https://www.gnu.org/software/gdb/bugs/>.
Find the GDB manual and other documentation resources online at:
    <http://www.gnu.org/software/gdb/documentation/>.

For help, type "help".
Type "apropos word" to search for commands related to "word"...
Reading symbols from ./lab_exam...
(gdb) break main
Breakpoint 1 at 0x1224: file lab_exam.c, line 19.
(gdb) run
Starting program: /home/bodike/Desktop/B24CS013_Bodike Chaithali_19/lab_exam
[Thread debugging using libthread_db enabled]
Using host libthread_db library "/lib/x86_64-linux-gnu/libthread_db.so.1".

Breakpoint 1, main () at lab_exam.c:19
19      int main() {
(gdb) next
20              int data[] = {4, 8, 15, 16, 23, 42};
(gdb) next
21              int n = sizeof(data) / sizeof(data[0]);
(gdb) set var data[0] = 100
(gdb) print data[0]
$1 = 100
(gdb) call sum_array(data, 6)
$2 = 204
(gdb) continue
Continuing.
Sum: 204
Max: 100
[Inferior 1 (process 15510) exited normally]
(gdb) exit
bodike@bodike-chaithali:~/Desktop/B24CS013_Bodike Chaithali_19$
```

Step 9:

```
bodike@bodike-chaithali:~/Desktop/B24CS013_Bodike Chaithali_19$ gdb ./lab_exam
GNU gdb (Ubuntu 12.1-0ubuntu1~22.04.2) 12.1
Copyright (C) 2022 Free Software Foundation, Inc.
License GPLv3+: GNU GPL version 3 or later <http://gnu.org/licenses/gpl.html>
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This GDB was configured as "x86_64-linux-gnu".
Type "show configuration" for configuration details.
For bug reporting instructions, please see:
<https://www.gnu.org/software/gdb/bugs/>.
Find the GDB manual and other documentation resources online at:
    <http://www.gnu.org/software/gdb/documentation/>.

For help, type "help".
Type "apropos word" to search for commands related to "word"...
Reading symbols from ./lab_exam...
(gdb) break sum_array
Breakpoint 1 at 0x1178: file lab_exam.c, line 4.
(gdb) run
Starting program: /home/bodike/Desktop/B24CS013_Bodike Chaithali_19/lab_exam
[Thread debugging using libthread_db enabled]
Using host libthread_db library "/lib/x86_64-linux-gnu/libthread_db.so.1".

Breakpoint 1, sum_array (arr=0x7fffffffe580, size=6) at lab_exam.c:4
4               int sum = 0;
(gdb) return 999
Make sum_array return now? (y or n) y
#0  0x0000555555555275 in main () at lab_exam.c:22
22              int total = sum_array(data, n);
(gdb) continue
Continuing.
Sum: 999
Max: 42
[Inferior 1 (process 15629) exited normally]
(gdb) exit
```

Step 10:

```
bodike@bodike-chaithali:~/Desktop/B24CS013_Bodike Chaithali_19$ gdb ./lab_exam
GNU gdb (Ubuntu 12.1-0ubuntu1~22.04.2) 12.1
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License GPLv3+: GNU GPL version 3 or later <http://gnu.org/licenses/gpl.html>
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This GDB was configured as "x86_64-linux-gnu".
Type "show configuration" for configuration details.
For bug reporting instructions, please see:
<https://www.gnu.org/software/gdb/bugs/>.
Find the GDB manual and other documentation resources online at:
    <http://www.gnu.org/software/gdb/documentation/>.

For help, type "help".
Type "apropos word" to search for commands related to "word"...
Reading symbols from ./lab_exam...
(gdb) break printf
Breakpoint 1 at 0x1070
(gdb) run
Starting program: /home/bodike/Desktop/B24CS013_Bodike Chaithali_19/lab_exam
[Thread debugging using libthread_db enabled]
Using host libthread_db library "/lib/x86_64-linux-gnu/libthread_db.so.1".

Breakpoint 1, 0x00007ffff7c606f0 in printf () from /lib/x86_64-linux-gnu/libc.so.6
(gdb) step
Single stepping until exit from function printf,
which has no line number information.
Sum: 108
main () at lab_exam.c:24
24              int maximum = max_element(data, n);
(gdb) exit
A debugging session is active.

        Inferior 1 [process 15904] will be killed.

Quit anyway? (y or n) y
bodike@bodike-chaithali:~/Desktop/B24CS013_Bodike Chaithali_19$
```

Question 3:

```
bodike@bodike-chaithali:~/Desktop/B24CS013_Bodike Chaithali_19$ vi dynamic_array_bug.c
bodike@bodike-chaithali:~/Desktop/B24CS013_Bodike Chaithali_19$ gcc -g -Wall -Wextra dynamic_array_bug.c -o dynamic_array_bug
bodike@bodike-chaithali:~/Desktop/B24CS013_Bodike Chaithali_19$ ./dynamic_array_bug
Array contents: 10 15 20 30 40
Array contents: 10 15 30 40
Array contents: 10 15 25 30 40
Insert Error: Invalid index or array full
Insert Error: Invalid index or array full
bodike@bodike-chaithali:~/Desktop/B24CS013_Bodike Chaithali_19$ gdb ./dynamic_array_bug
GNU gdb (Ubuntu 12.1-0ubuntu1~22.04.2) 12.1
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This GDB was configured as "x86_64-linux-gnu".
Type "show configuration" for configuration details.
For bug reporting instructions, please see:
<https://www.gnu.org/software/gdb/bugs/>.
Find the GDB manual and other documentation resources online at:
    <http://www.gnu.org/software/gdb/documentation/>.

For help, type "help".
Type "apropos word" to search for commands related to "word"...
Reading symbols from ./dynamic_array_bug...
(gdb) break main
Breakpoint 1 at 0x13f7: file dynamic_array_bug.c, line 46.
(gdb) break insert
Breakpoint 2 at 0x123c: file dynamic_array_bug.c, line 19.
(gdb) break removeAt
Breakpoint 3 at 0x12f6: file dynamic_array_bug.c, line 30.
(gdb) run
Starting program: /home/bodike/Desktop/B24CS013_Bodike Chaithali_19/dynamic_array_bug
[Thread debugging using libthread_db enabled]
Using host libthread_db library "/lib/x86_64-linux-gnu/libthread_db.so.1".

Breakpoint 1, main () at dynamic_array_bug.c:46
46      int main() {
```

```
46      int main() {
(gdb) next
48              initArray(&arr);
(gdb) next
49              insert(&arr, 0, 10);
(gdb) tep
Undefined command: "tep".  Try "help".
(gdb) step

Breakpoint 2, insert (arr=0x7fffffffe570, index=0, value=10) at dynamic_array_bug.c:19
19              if (index < 0 || index > arr->size || arr->size == arr->capacity) {
(gdb) display arr->size
1: arr->size = 0
(gdb) display arr->capacity
2: arr->capacity = 5
(gdb) display i
No symbol "i" in current context.
(gdb) next
23              for (int i = arr->size; i >= index; i--)
1: arr->size = 0
2: arr->capacity = 5
(gdb) display i
3: i = 32767
(gdb) next
24                      arr->data[i] = arr->data[i-1];
1: arr->size = 0
2: arr->capacity = 5
3: i = 0
(gdb) display i
4: i = 0
(gdb) next
23              for (int i = arr->size; i >= index; i--)
1: arr->size = 0
2: arr->capacity = 5
3: i = 0
4: i = 0
(gdb) continue
Continuing.
```

```
Breakpoint 2, insert (arr=0x7fffffffe570, index=1, value=20) at dynamic_array_bug.c:19
19              if (index < 0 || index > arr->size || arr->size == arr->capacity) {
1: arr->size = 1
2: arr->capacity = 5
(gdb) next
23              for (int i = arr->size; i >= index; i--)
1: arr->size = 1
2: arr->capacity = 5
3: i = -1
4: i = -1
(gdb) next
24                      arr->data[i] = arr->data[i-1];
1: arr->size = 1
2: arr->capacity = 5
3: i = 1
4: i = 1
(gdb) next
23              for (int i = arr->size; i >= index; i--)
1: arr->size = 1
2: arr->capacity = 5
3: i = 1
4: i = 1
(gdb) continue
Continuing.

Breakpoint 2, insert (arr=0x7fffffffe570, index=2, value=30) at dynamic_array_bug.c:19
19              if (index < 0 || index > arr->size || arr->size == arr->capacity) {
1: arr->size = 2
2: arr->capacity = 5
(gdb) next
23              for (int i = arr->size; i >= index; i--)
1: arr->size = 2
2: arr->capacity = 5
3: i = 0
4: i = 0
(gdb) next
24                      arr->data[i] = arr->data[i-1];
1: arr->size = 2
```

```
24                      arr->data[i] = arr->data[i-1];
1: arr->size = 2
2: arr->capacity = 5
3: i = 2
4: i = 2
(gdb) next
23              for (int i = arr->size; i >= index; i--)
1: arr->size = 2
2: arr->capacity = 5
3: i = 2
4: i = 2
(gdb) next
25              arr->data[index] = value;
1: arr->size = 2
2: arr->capacity = 5
(gdb) continue
Continuing.

Breakpoint 2, insert (arr=0x7fffffffe570, index=1, value=15) at dynamic_array_bug.c:19
19              if (index < 0 || index > arr->size || arr->size == arr->capacity) {
1: arr->size = 3
2: arr->capacity = 5
(gdb) next
23              for (int i = arr->size; i >= index; i--)
1: arr->size = 3
2: arr->capacity = 5
3: i = 1
4: i = 1
(gdb) next
24                      arr->data[i] = arr->data[i-1];
1: arr->size = 3
2: arr->capacity = 5
3: i = 3
4: i = 3
(gdb) next
23              for (int i = arr->size; i >= index; i--)
1: arr->size = 3
2: arr->capacity = 5
3: i = 3
```

```
3: i = 3
4: i = 3
(gdb) next
24                      arr->data[i] = arr->data[i-1];
1: arr->size = 3
2: arr->capacity = 5
3: i = 2
4: i = 2
(gdb) next
23              for (int i = arr->size; i >= index; i--)
1: arr->size = 3
2: arr->capacity = 5
3: i = 2
4: i = 2
(gdb) next
24                      arr->data[i] = arr->data[i-1];
1: arr->size = 3
2: arr->capacity = 5
3: i = 1
4: i = 1
(gdb) next
23              for (int i = arr->size; i >= index; i--)
1: arr->size = 3
2: arr->capacity = 5
3: i = 1
4: i = 1
(gdb) continue
Continuing.

Breakpoint 2, insert (arr=0x7fffffffe570, index=4, value=40) at dynamic_array_bug.c:19
19              if (index < 0 || index > arr->size || arr->size == arr->capacity) {
1: arr->size = 4
2: arr->capacity = 5
(gdb) next
23              for (int i = arr->size; i >= index; i--)
1: arr->size = 4
2: arr->capacity = 5
3: i = 0
```

```
3: i = 0
4: i = 0
(gdb) next
24                      arr->data[i] = arr->data[i-1];
1: arr->size = 4
2: arr->capacity = 5
3: i = 4
4: i = 4
(gdb) next
23              for (int i = arr->size; i >= index; i--)
1: arr->size = 4
2: arr->capacity = 5
3: i = 4
4: i = 4
(gdb) next
25              arr->data[index] = value;
1: arr->size = 4
2: arr->capacity = 5
(gdb) continue
Continuing.
Array contents: 10 15 20 30 40

Breakpoint 3, removeAt (arr=0x7fffffffe570, index=2) at dynamic_array_bug.c:30
30              if (index < 0 || index >= arr->size) {
(gdb) next
34              for (int i = index; i < arr->size - 1; i++)
(gdb) display i
5: i = 5
(gdb) next
35                      arr->data[i] = arr->data[i+1];
5: i = 2
(gdb) next
34              for (int i = index; i < arr->size - 1; i++)
5: i = 2
(gdb) next
35                      arr->data[i] = arr->data[i+1];
5: i = 3
(gdb) next
```

```
(gdb) next
34              for (int i = index; i < arr->size - 1; i++)
5: i = 3
(gdb) next
36              arr->size--;
(gdb) next
37      }
(gdb) continue
Continuing.
Array contents: 10 15 30 40

Breakpoint 2, insert (arr=0x7fffffffe570, index=2, value=25) at dynamic_array_bug.c:19
19              if (index < 0 || index > arr->size || arr->size == arr->capacity) {
1: arr->size = 4
2: arr->capacity = 5
(gdb) next
23              for (int i = arr->size; i >= index; i--)
1: arr->size = 4
2: arr->capacity = 5
3: i = 4
4: i = 4
(gdb) continue
Continuing.
Array contents: 10 15 25 30 40

Breakpoint 2, insert (arr=0x7fffffffe570, index=5, value=50) at dynamic_array_bug.c:19
19              if (index < 0 || index > arr->size || arr->size == arr->capacity) {
1: arr->size = 5
2: arr->capacity = 5
(gdb) next
20                      printf("Insert Error: Invalid index or array full\n");
1: arr->size = 5
2: arr->capacity = 5
(gdb) next
Insert Error: Invalid index or array full
21                      return;
1: arr->size = 5
2: arr->capacity = 5
```

```
1: arr->size = 5
2: arr->capacity = 5
(gdb) next
27      }
1: arr->size = 5
2: arr->capacity = 5
(gdb) next
main () at dynamic_array_bug.c:60
60              insert(&arr, -1, 60);
(gdb) next
continue
No symbol "continue" in current context.
(gdb) next

Breakpoint 2, insert (arr=0x7fffffffe570, index=-1, value=60) at dynamic_array_bug.c:19
19              if (index < 0 || index > arr->size || arr->size == arr->capacity) {
1: arr->size = 5
2: arr->capacity = 5
(gdb) continue
Continuing.
Insert Error: Invalid index or array full
[Inferior 1 (process 19417) exited normally]
(gdb) quit
bodike@bodike-chaithali:~/Desktop/B24CS013_Bodike Chaithali_19$ vi dynamic_array_bug.c
bodike@bodike-chaithali:~/Desktop/B24CS013_Bodike Chaithali_19$ gcc -g -Wall -Wextra dynamic_array_bug.c -o dynamic_array_fixed
bodike@bodike-chaithali:~/Desktop/B24CS013_Bodike Chaithali_19$ ./dynamic_array_fixed
Array contents: 10 15 20 30 40
Array contents: 10 15 30 40
Array contents: 10 15 25 30 40
Insert Error: Invalid index or array full
Insert Error: Invalid index or array full
bodike@bodike-chaithali:~/Desktop/B24CS013_Bodike Chaithali_19$ gdb ./dynamic_array_fixed
GNU gdb (Ubuntu 12.1-0ubuntu1~22.04.2) 12.1
Copyright (C) 2022 Free Software Foundation, Inc.
License GPLv3+: GNU GPL version 3 or later <http://gnu.org/licenses/gpl.html>
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.
Type "show copying" and "show warranty" for details.
```

```
Type "show copying" and "show warranty" for details.
This GDB was configured as "x86_64-linux-gnu".
Type "show configuration" for configuration details.
For bug reporting instructions, please see:
<https://www.gnu.org/software/gdb/bugs/>.
Find the GDB manual and other documentation resources online at:
    <http://www.gnu.org/software/gdb/documentation/>.

For help, type "help".
Type "apropos word" to search for commands related to "word"...
Reading symbols from ./dynamic_array_fixed...
(gdb) break main
break insert
break removeAt
Function "main
break insert
break removeAt" not defined.
Make breakpoint pending on future shared library load? (y or [n]) n
(gdb) break main
Breakpoint 1 at 0x13fa: file dynamic_array_bug.c, line 47.
(gdb) break insert
Breakpoint 2 at 0x123c: file dynamic_array_bug.c, line 19.
(gdb) break removeAt
Breakpoint 3 at 0x1279: file dynamic_array_bug.c, line 31.
(gdb) run
Starting program: /home/bodike/Desktop/B24CS013_Bodike Chaithali_19/dynamic_array_fixed
[Thread debugging using libthread_db enabled]
Using host libthread_db library "/lib/x86_64-linux-gnu/libthread_db.so.1".

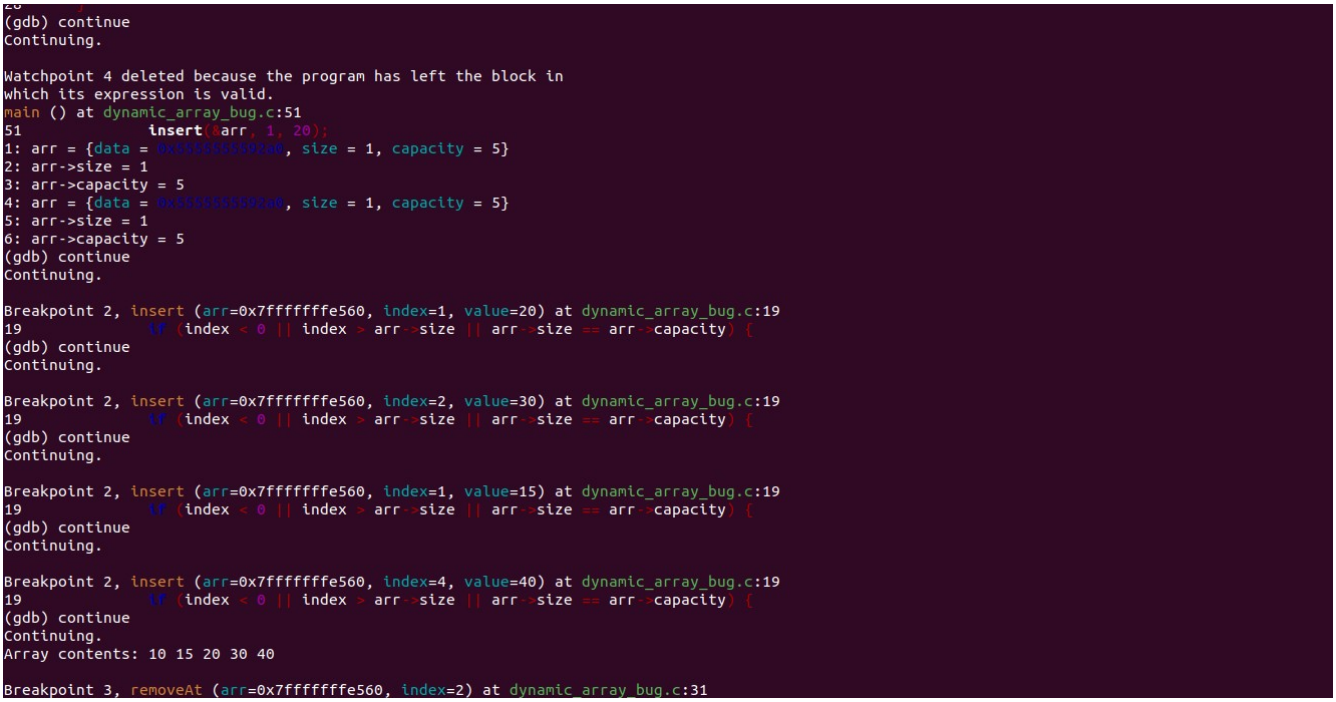
Breakpoint 1, main () at dynamic_array_bug.c:47
47      int main() {
(gdb) display arr
1: arr = {data = 0x7fffffffe000, size = 100, capacity = 0}
(gdb) display arr->size
2: arr->size = 100
(gdb) display arr->capacity
3: arr->capacity = 0
(gdb) next
49              initArray(&arr);
```

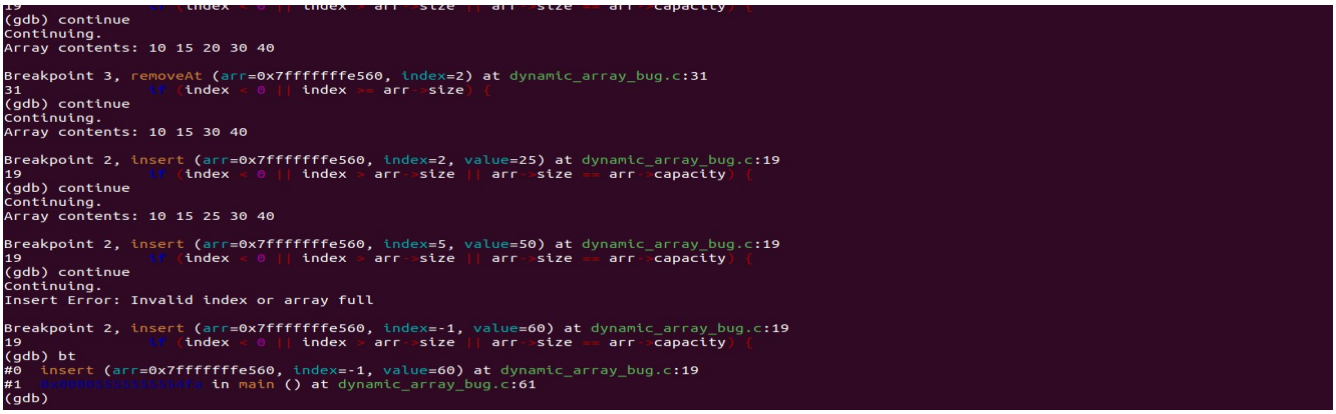
```
(gdb) next
49              initArray(&arr);
1: arr = {data = 0x7fffffffe000, size = 100, capacity = 0}
2: arr->size = 100
3: arr->capacity = 0
(gdb) display arr
4: arr = {data = 0x7fffffffe000, size = 100, capacity = 0}
(gdb) display arr->size
5: arr->size = 100
(gdb) display arr->capacity
6: arr->capacity = 0
(gdb) next
50              insert(&arr, 0, 10);
1: arr = {data = 0x555555559260, size = 0, capacity = 5}
2: arr->size = 0
3: arr->capacity = 5
4: arr = {data = 0x555555559260, size = 0, capacity = 5}
5: arr->size = 0
6: arr->capacity = 5
(gdb) step

Breakpoint 2, insert (arr=0x7fffffffe560, index=0, value=10) at dynamic_array_bug.c:19
19              if (index < 0 || index > arr->size || arr->size == arr->capacity) {
(gdb) watch arr.size
Hardware watchpoint 4: arr.size
(gdb) next
24              for (int i = arr->size - 1; i >= index; i--)
(gdb) next
26              arr->data[index] = value;
(gdb) continue
Continuing.

Hardware watchpoint 4: arr.size

Old value = 0
New value = 1
insert (arr=0x7fffffffe560, index=0, value=10) at dynamic_array_bug.c:28
28      }
(gdb) continue
```





changed apt of the code:

```
void insert(DynamicArray* arr, int index, int value) {
    if (index < 0 || index > arr->size || arr->size == arr->capacity) {
        printf("Insert Error: Invalid index or array full\n");
        return;
    }

    // FIX: Shift elements correctly (start from arr->size-1)
    for (int i = arr->size - 1; i >= index; i--) {
```

```
        arr->data[i + 1] = arr->data[i];
    }


    arr->data[index] = value;
    arr->size++;
}
```

I used **breakpoints (b)** at `main`, `insert`, and `removeAt` to pause execution at important points and observe how the array changed after every operation.

I set a **watchpoint** on `arr.size` to detect unexpected modification of the size variable during insertions and removals.

While stepping through the code, I found that the program was accessing memory out of bounds during the shift operation inside `insert()`.